Data Structures and Algorithms

Project Proposal

Document Verification System

Section: A

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Submitted To:

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Problem Statement:

Managing appointments and verifying documents can be a slow and error-prone process. Many systems today still struggle with delays, lack of personalization, and manual document checks, which cause a lot of issues for both users and admins. This project aims to build a system that makes appointment scheduling, document verification, and status tracking easier, faster, and more reliable. This innovation demonstrates the potential for technology to streamline essential processes, ultimately saving time and resources.

Significance and Relevance:

In many industries like education or visa processing, managing appointments and verifying documents is essential. By using simple data structures like **linked lists**, **queues**, **stacks**, **trees**, **graphs** and different **sorting algorithms**, we can build a system that is fast, secure, and easy to use. This project will show how these concepts can be used to solve real-world problems and improve the user experience.

Features

- **User Account Management**: Users can securely create and manage accounts, log in, and update their profiles, ensuring a safe and personalized experience.
- **Appointment Booking System**: Applicants can view and book available appointment slots using Binary Search Trees (BST), providing efficient slot management and reduced booking conflicts.
- **Priority Appointment Handling** Urgent cases are prioritized using priority queues, allowing quick processing for time-sensitive scenarios such as medical or emergency appointments.
- **Document Verification**: Applicants can upload PDF documents for verification. Admins review and validate the documents, with the system flagging any errors or missing files.
- **Status Tracking**: The system manages and visualizes appointment status transitions (e.g., *Booked → In Process → Verified/Rejected*). Using **graph-based models**, it ensures clear and traceable progress tracking, reducing confusion and errors for both users and admins.
- **Admin Controls**: Admins have access to comprehensive tools for managing appointments, updating statuses, and verifying documents. The system includes a **confirmation step** before applying critical changes (e.g., status updates or document rejections), reducing the risk of accidental errors.

Members and their Contributions:

- **Syed Shayan Arshad:** Responsible for implementing admin functionalities, including appointment management and document verification workflows
- **Hammad Khalid:** Focused on developing user functionalities, including secure account management and appointment booking.
- **Adeel Ul Rehman:** Managed the status tracking system, ensuring smooth transitions through stages and efficient handling of appointment progress and document verification.